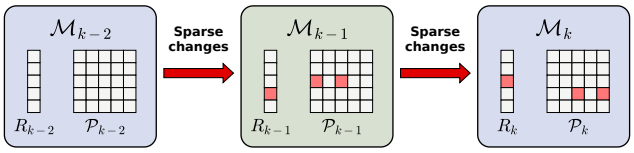
Adaptive Memory for Online Reinforcement Learning under Sparse Model Changes

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Problem and method. Each deployment supplies H observations from a fixed MDP $\mathcal{M}_k = (\mathcal{S}, \mathcal{A}, P_k, R_k, \gamma)$. At each known boundary, at most ϵd reward entries and ϵnd transition entries change, with unknown locations and magnitudes. The learner selects separate histories for every reward entry and transition row. It reuses the longest history whose nested confidence sets have a nonempty intersection, fits the model within these intersections, and plans optimally. It stores summaries for at most B completed deployments.



Sparse reward and transition changes across deployments.

Assumptions and notation. Let $n = |\mathcal{S}|$, $d = n|\mathcal{A}|$, and $\vartheta = \min\{1, n\epsilon/2\}$, so at most ϑd transition rows change. Assume $|r| \le \overline{R}$, conditionally independent tuples given an MDP sequence independent of observation noise, sampling coverage $\nu_k(s,a) \ge v > 0$, and discounted occupancies $d^{\pi}_{\rho,k}(s,a) \le \kappa/d$. For discounted return $J^{\rho}_k(\pi)$ from a fixed ρ, define

$$\mathcal{L}^{\rho}_{K,H} := \frac{1}{KH}\sum_{k=0}^{K-1}\sum_{h=0}^{H-1}\left(J^{\rho,*}_k - J^{\rho}_k(\pi_{k,h})\right), \qquad \mathcal{R}^{\rho}_{K,H} := \mathbb{E}[\mathcal{L}^{\rho}_{K,H}],$$
$$L := \log\frac{8d(B+1)(H+1)}{\delta}, \qquad \Lambda := n\log 2 + L, \qquad \beta_{K,B} := K^{-1} + B^{-1}.$$

1. Finite-deployment policy loss. For every $K \ge 1$, integer $B \ge 1$, and $\delta \in (0,1)$,

$$\mathcal{R}^{\rho}_{K,H} \lesssim \frac{\overline{R}}{(1-\gamma)^2}\left[\sqrt{\frac{L(\beta_{K,B}+\kappa\epsilon)}{vH}} + \sqrt{\frac{\Lambda(\beta_{K,B}+\kappa\vartheta)}{vH}}\right] + \frac{\overline{R}\delta}{1-\gamma}. \tag{1}$$

2. High-probability guarantee. Fix K and $\alpha \in (0,1)$, and choose $\delta = \alpha/K$. With $L_\alpha = \log(8dK(B+1)(H+1)/\alpha)$ and $\Lambda_\alpha = n\log 2 + L_\alpha$, with probability at least $1-\alpha$,

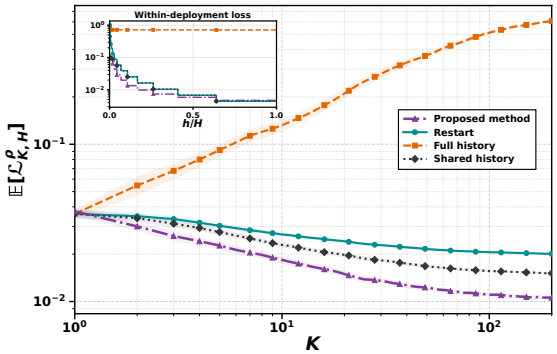
$$\mathcal{L}^{\rho}_{K,H} \lesssim \frac{\overline{R}}{1-\gamma}\sqrt{\frac{L_\alpha(\beta_{K,B}+\kappa\epsilon)}{vH}} + \frac{\gamma\overline{R}}{(1-\gamma)^2}\sqrt{\frac{\Lambda_\alpha(\beta_{K,B}+\kappa\vartheta)}{vH}}. \tag{2}$$

3. Persistent tracking with bounded memory. For $0 < \epsilon \le 1$, choose $B = \lceil \epsilon^{-1} \rceil$ and $\delta = \epsilon/(H+1)$, independently of K. Let $L_\epsilon = \log(24d(H+1)^2/\epsilon^2)$ and $\Lambda_\epsilon = n\log 2 + L_\epsilon$. Then

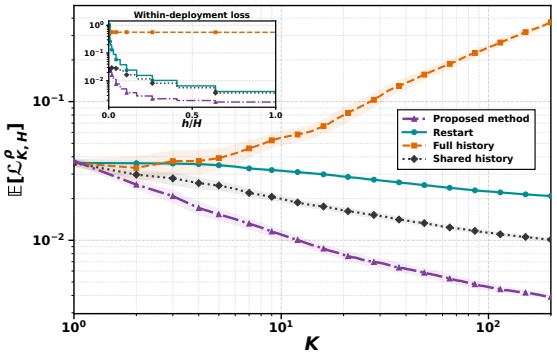
$$\mathcal{J}^{\rho}_H := \limsup_{K\to\infty} \mathcal{R}^{\rho}_{K,H} \lesssim \frac{\overline{R}}{1-\gamma}\sqrt{\frac{\kappa}{vH}}\left[\sqrt{\epsilon L_\epsilon} + \frac{\gamma\sqrt{\vartheta\Lambda_\epsilon}}{1-\gamma}\right]. \tag{3}$$

When $0 < n\epsilon \le 2$, $\mathcal{J}^{\rho}_H = \widetilde{O}(\sqrt{\epsilon/H})$ for fixed remaining parameters. Only $O(\epsilon^{-1})$ deployment summaries are retained, using $O(n^2|\mathcal{A}|/\epsilon)$ scalar storage.

Numerical results. $n = 50$, $|\mathcal{A}| = 4$, $\gamma = 0.8$, $H = 2^{20}$, $K = 200$, and $B = 100$. Results average 50 independent runs. Within each run, all methods receive identical observations and use a common geometric policy-update grid. Reported reductions compare average policy loss over the final 40 deployments against shared history.



(a) $\epsilon = 0.02$: **28.2%** lower policy loss.



(b) $\epsilon = 0.005$: **65.2%** lower policy loss.